

NAMAN SHARMA

Robotics Engineer | Computer Vision | ROS 2 | Edge AI

naman4015@gmail.com | +91 84378 88862 | Mohali, Punjab, India |

<https://naman2328.github.io/naman-robotics-vision-portfolio/index.html> | [linkedin.com/in/namansharma4015](https://www.linkedin.com/in/namansharma4015)

PROFESSIONAL SUMMARY

Results-driven Mechatronics Engineering graduate with 12 months of high-impact experience in production environments at a Tier-1 automotive supplier. Expert in designing and deploying real-world edge AI and computer vision solutions using NVIDIA Jetson and ROS 2. Proven ability to optimize inference pipelines for latency and reliability while collaborating across cross-functional teams in ISO-certified facilities.

WORK EXPERIENCE

Process Engineering Intern — Autoliv India — Tier-1 Automotive Supplier *Feb 2025 – Feb 2026*

ISO-Certified Facility · Jhajjar, India

- Engineered and deployed a high-accuracy (95%) vision-based OCR inspection system on NVIDIA Jetson Orin Nano Super, reducing inference latency to 150–250 ms in live production environments.
- Developed a robust industrial thread-detection pipeline utilizing color-based ROI classification, which decreased manual inspection time by 40%.
- Optimized real-time inference pipelines for high-throughput edge deployment, ensuring stable thermal performance and zero cloud dependency.
- Designed an intuitive Qt GUI for QC operators, streamlining OK/NG decision logging and audit trails, achieving immediate team adoption without additional training.
- Maintained strict adherence to ISO quality standards while collaborating with mechanical and process engineering teams to deliver auditable, reliable production documentation.

PROJECTS

Vision-Guided Pick & Place System | *ROS 2 · OpenCV · Python · Dobot Magician*

- Architected a decoupled ROS 2 system with separate nodes for perception, planning, and execution, enabling independent pipeline upgrades without modifying control logic.
- Implemented 2D camera-to-robot coordinate transformation for precise object grasping, achieving ± 2 mm positional repeatability on physical Dobot hardware.
- Eliminated servo jitter through systematic PID gain tuning, improving motion consistency across repeated grasping cycles.

Industrial OCR Inspection System — Edge Deployment | *PaddleOCR · OpenCV · Python · Jetson Orin Nano Super · Qt*

- Built a ROI-based inspection pipeline with confidence thresholding for deterministic OK/NG classification, handling glare and inconsistent part positioning in real factory environments.
- Achieved full offline deployment on Jetson Orin Nano Super with real industrial camera integration, production UI, result logging, and audit trail, eliminating cloud dependency.

Omnidirectional Mecanum Mobile Robot | *Python · ROS 2 · Motor Control · Mecanum Kinematics · Embedded Systems*

- Built a holonomic mobile robot with full kinematic model enabling lateral translation, diagonal motion, and in-place rotation without chassis rotation.
- Implemented per-wheel velocity vector control; platform is ROS 2-ready for SLAM and autonomous navigation integration.

Custom Multi-DOF Robotic Arm | *Python · Servo Motors · 3D Printing · PWM Calibration · Mechanical Design*

- Designed, 3D-printed, and assembled a multi-DOF robotic arm from scratch, closing the full hardware-software integration loop from mechanical design to PWM-calibrated servo control.
- Programmed repeatable joint trajectory sequences and resolved real-world backlash and torque-limit constraints not present in simulation.

TECHNICAL SKILLS

Robotics: ROS 2 (Humble), SLAM, Sensor Fusion, Motion Planning, URDF/Xacro, Robot Kinematics, LiDAR, Encoders

Computer Vision: OpenCV, PaddleOCR, ROI Pipelines, Object Detection, Image Preprocessing, OK/NG Classification

Edge AI & Deployment: NVIDIA Jetson Orin Nano Super, TensorRT, ONNX, Real-Time Inference Optimization

Simulation: Gazebo (Fortress/Classic), Isaac Sim, RViz

Programming Languages: Python (primary), C++ (working proficiency), Bash

Hardware: NVIDIA Jetson Orin Nano, Raspberry Pi, Arduino, Dobot Magician, Mecanum Drive, Servo/Motor Control

Tools & Process: Git, Linux (Ubuntu), Qt, ISO-Compliant Documentation, Deterministic System Design

EDUCATION

B.E. Mechatronics Engineering 2022 – 2026

Chitkara University, Punjab, India

Relevant Coursework: Robotics, Control Systems, Embedded Systems, Machine Vision, Mechatronics System Design